

1 Getting Started

1.1 Introduction

Welcome to M-Pesa's Developer Portal! The OpenAPI Solution offers RESTful interfaces to help you create a payment solution that meets merchant and business needs. These interfaces enable businesses to incorporate Mobile Money payment functionality into newly developed applications and point of sale devices. In this web portal, detailed documentation will be provided on how to build and test interfaces against our RESTful API endpoints. The API endpoints that have been exposed allows for:

- customer to business (C2B) single payment
- customer to business (C2B) two-step payment
- business to customer (B2C)
- business to business (B2B)
- payment reversals
- query transaction status
- direct debit creation and payments
- query beneficiary name (QBN)
- query direct debit
- cancel direct debit

1.2 Development Process

The M-Pesa Developer Portal allows developers the following functions:

- Application Management - View, edit and create new integration services through applications by selecting products, defining their scope and setting limits
- Rapid Integration - Develop interfaces quickly by following detailed API documentation, examples and library codes
- Sandbox Testing - Secure and isolated testing environment to test interfaces and confirm responses through pre-determined scenarios
- Developer and Organisation Linking – Connect with organisations and build new applications to integrate with Mobile Money
- Push for Review Workflow – Initiate a seamless workflow to hand over integration configuration to linked organisations for testing and go-live
- Integrated Support Forum - Get help from experts in the field on various topics
- Dedicated Business Processes - Understand the flow of interfaces / interactions to build better products and inspire future integration services

1.3 Developing your API

1.3.1 Overview

The APIs that have been exposed are clearly documented in their respective sections within the API DOCUMENTATION pages. Following along with the structure and parameters will allow you to rapidly build the correct API package to seamlessly integrate with Mobile Money services, allowing for convenient payment opportunities for businesses.

1.3.2 RESTful APIs

The APIs exposed are RESTful APIs that break down transactions into a series of small modules, each addressing an underlying part of the transaction. The REST APIs provide developers with all the tools to effectively interact with the Mobile Money Payments Gateway by calling specific functions using the API. The first functional API call to be initiated will be to generate a Session Key which allows the calling application to be authorised and authenticated in the OpenAPI server. If successful, the developer will receive a valid session key. Using the session key, subsequent API calls can be made to initiate the payment transaction needed (business to customer, customer to business, etc.). The session key will only be valid for a finite lifetime and is configured within the Applications page.

1.4 Applications

Developers can manage applications and their respective business configuration on the APPLICATIONS page. Each application is a containerised record that lists the integration configuration, product scope that could be called and any/all limitations placed on the API calls. The developer can create new applications, duplicate existing configuration into a new application, edit or remove existing applications.

Screen for creating a new application

The screenshot shows the 'APPLICATION DETAILS' form. It has a red header bar with the title '< APPLICATION DETAILS'. The form is divided into several sections:

- Application Name *** and **Version *** input fields.
- Description** section with a text area containing 'Sandbox API Key', 'Auto-Generated', and 'Session Lifetime *'.
- Status** dropdown menu with 'In Development' selected.
- Units** dropdown menu with 'Hours' selected.
- Trusted Sources *** section with a text area for 'Comma separated list of allowed IPs (wildcards are allowed)'.
- Currencies** dropdown menu.
- Payment Methods** section with buttons: B2B SINGLE PAYMENT (highlighted in red), B2C SINGLE PAYMENT, C2B SINGLE PAYMENT, C2B MULTISTAGE PAYMENT, DIRECT DEBIT, QUERY BENEFICIARY NAME, QUERY DIRECT DEBIT, CANCEL DIRECT DEBIT, REVERSAL, TX STATUS QUERY, and TX STATUS UPDATE.
- B2B Single Payment** section with a toggle switch and several checkboxes: Asynchronous Flow, Limit API Usage Time, Limit Single Transaction Value, Limit Session Transaction Value, Limit Session Number of Transactions, Limit Customer Session Transaction Value, and Limit Customer Session Number of Transactions.
- CANCEL** and **SAVE** buttons at the bottom right.

Applications require the following details:

- Application Name – human-readable name of the application
- Version – version number of the application, allowing changes in API products to be managed in different versions
- Description – Free text field to describe the use of the application
- API Key – Unique authorisation key used to authenticate the application on the first call. API Keys need to be encrypted in the first “Generate Session API Call” to create a valid session key to be used as an access token for future calls. Encrypting the API Key is documented in the GENERATE SESSION API page

- Session Lifetime – The session key has a finite lifetime of availability that can be configured. Once a session key has expired, the session is no longer usable, and the caller will need to authenticate again.
- Trusted Sources – the originating caller can be limited to specific IP address(es) as an additional security measure.
- Products / Scope / Limits – the required API products for the application can be enabled and limits defined for each call.

1.5 API Products

Developers will have access to all API products within the OpenAPI. Each API call needed should be enabled using the toggle. Once enabled, the user can make use of the API call. Businesses will be required to have the product enabled on their M-Pesa account. More details of the API product are listed in the API Documentation pages. The API scope can be limited by enabling and setting the values on a per API basis. Limits for the API is enforced per session

- Usage time – which time of the day the API may be used
- Single transaction value – value of a single transaction in a given currency
- Daily transaction value – accumulated value of all transactions for the day in a given currency
- Number of transactions – accumulated number of transactions for the day Once the session limits are reached for the API, no further transaction can be completed until a new session key is generated. In addition to session limits, the APIs can be limited on a customer basis by configuring the following limits:
- Customer daily transaction value – accumulated value of all transactions for a specific customer during the day in a given currency
- Customer number of transactions – accumulated number of transactions initiated by a specific transaction for the day

1.6 Testing within Sandbox

1.6.1 Overview

New and existing applications can be safely tested within the dedicated Sandbox environment. The Sandbox server isolate transaction testing away from the production Mobile Money servers for safe-guarding purposes. The developer can trigger pre-built scenarios to test the following use cases:

- Successful payments
- Rejected payments
- Timeout on payments
- Insufficient balance
- Failed transactions

In addition to pre-built scenarios being triggered, the developer can register test mobile handset numbers (MSISDNs) to trigger USSD Push transactions for C2B payments. Detailed information on testing can be found in the TESTING DOCUMENTATION page.

1.7 Linking with Businesses / Organisations

1.7.1 Overview

A developer account needs to be linked to a business/organisation before an application can be taken live. Developer accounts can create new applications and set the initial configuration. An Organisation is registered on the Mobile Money platform. After the commercial agreement with the Mobile Network Operator (MNO), the organisation user will be able to promote a new application to production.

1.7.2 Linking Process

To link a developer account with an organisation, the organisation user will send an invitation to the developer's e-mail account from within the Portal. If the developer user already has an account, a notification will be sent to the Developer user to accept the new link. If the developer does not yet have an account, an e-mail will be sent to the Developer user's e-mail account with a link to register onto the Developer Portal. Once the account is created, the developer and the organisation will be linked. The developer will be able to develop on behalf of the organisation.

1.8 Handing over for Review

1.8.1 Overview

The developer is responsible for creating and developing integration on behalf of the organisation. Once the development of the integration is completed and testing is concluded, the developer user can hand over the created application to the linked organisation through the Developer Portal.

1.8.2 Send for Review

The developer can only hand over an application to an organisation with whom a link has been established (developer has accepted the invitation of the organisation). From the APPLICATION page, the developer can click on the application and click "Send for Review". The developer can select the organisation to whom he/she wants to hand over the application and its configuration. The OpenAPI workflow will move the application to the organisation user's account who is able to review the application configuration, scope and limits. Once reviewed, the application can be accepted or rejected. Once accepted the application and its scope will be locked for the developer. The developer user will not be able to do any changes on the application.

1.9 Support and Documentation

The OpenAPI offers several sections of support and documentation to get help from experts in the field on various topics.

1.10 Documentation

Dedicated documentation pages exist to assist in the building of the API interfaces. How to build each API call, including the required parameters for the header and body, and the expected response/result fields are documented in detail. Testing documentation exists to guide the user on how to trigger specific responses to test the application's interfaces. In addition, each page within the portal consists of a HELP page to provide relevant information, guiding the user what needs to be done at each section.

1.10.1 Support Forum

In addition to the documentation pages, a support forum exists to ask questions and seek help from other developers and operational support teams.

2 API Documentation

2.1 Generate SessionKey

Before you can integrate on the M-Pesa OpenAPI solution, you must exchange your Application Key for a Session Key. The API Key is created with the creation of a new application. The Session Key acts as an access token that authorises the rest of your REST API calls to the system. A valid Session Key is needed to transact on M-Pesa using OpenAPI.

2.1.1 Generating the Session Key

2.1.1.1 Encrypting the API Key

To generate your Session Key for the sandbox and live environments: 1. Log into OpenAPI with a developer account 2. On the APPLICATION page, click Create New Application. Creating an application will generate your unique API Application Key needed to authorise and authenticate your application on the server 3. Type your name and version number for the application and select the products you wish to use. (The application can be configured any time). Save your new application. 4. Copy and save the API Key. 5. Copy the Public Key from the below section. 6. Generate a decoded Base64 string from the Public Key 7. Generate an instance of an RSA cipher and use the Base 64 string as the input 8. Encode the API Key with the RSA cipher and digest as Base64 string format 9. The result is your encrypted API Key.

2.1.1.2 Platform Public Key

Use the Public Key associated to the targeted platform to encrypt your API Key:

```
MIICijANBgkqhkiG9w0BAQEFAAOCAg8AMIICCgKCAgEArv9yxA69XQKBo24BaF/D+fv1qmGdYjqLQ5WtNBb5tquqGvAvG3WMFETVUSow/LizQalxj2ElMVrUmzu5mGGkxK08bWEXF7a1DEvtVJs6npI1fJc2SnrU14AOrIrB28ogm58JjAl5BOQawOXD5dfSk7MaAA82pVHoIqEu0Fx8BOKU+RGTiHrU+ptwlj4bsAJYiPbSX6i71gfPvwHPYamM0bfI4CmlsUUR3KvCG24rB6FNPcRBhM3jDuv8ae2kC33w9hEq8qNB55uw51vK7hyXoAa+U7IqPly6nBdlN25gkxEA8yrsl1678cspeXr+3ciRyqoRgj9RD/ONbJhhxvFvt1cLBh+qwK2eqISfBb06eRnNeC71oBokDm3zyCnkOtMDG17IvnMfZfEPFCfg5QgJVklmsPpRvQxmEsRvX9MQRYFVzgy2CWNlb7c+jPapyrNwoUbAN1N8adU1m6yOuoX7F49x+OjiG2se0EJ6nafekUXw/+hiJZvELUYgzKUtMAZVTNZfT8jjb58j8GVtuS+6TM2AutbejaCV84ZK58E2CRJghmjQibEU06KPdD7oTlEkFy52Y1uOOBXgYpqMzuFNpmdqggSM4dU70PO8ogyKGI LAIXCetMjjm6FCMEA3Kc8K0Ig7/XtFm9By6VxTJK1Mg36TlHaZKP6VzVlXMtesJECawEAAQ==
```

2.1.1.3 API Header Parameters

The encrypted API Key needs to be added to the Header block that makes up your initial Generate SessionKey API Call in the Authorization-Bearer value. The rest of the header parameters are as follows:

Header Name	Header Value	Header Description
Content-Type	application/json	The content-type of the API

Authorization	Bearer [encrypted-APIKey]	Bearer token authorization is used. The encrypted APIKey needs to be used.
Origin	*	Used to limit the origin of the API caller to specific systems. Use the domain name or IP address. Should match the value(s) defined in your Application.

2.1.1.4 Sample Header Code

Content-Type: application/json Authorization: Bearer K2sf0ne6lAocJqWx03kTHM BmLl0fQyBQWV7GfH3z1kMxV qMCB7sMXn+CKigIPIawHoNr2IrGdjBdFugyHNxRnGkrfSDkC9PF BhR/D9ePdwl5jgggnG8FItGpw4TEwECgzk0sEmYmKaB9K6lgvs4rs7TMPpTom7uX72+tZ+qMqRV t7LrGsngBrkXjb0xc19fkImVHjehzjVCOSOsROsx4nRlM9wtl4KSLgKCBzmLePRmwu/IOCF3NrI QLVYYqUDDSGYbHURk1L6d4kxJffFM/wa2BPoeuG59gcglDyEaQ5Sv5VnaGKDh1XTBrvKS+J9Efw g7s1CWLQxhtYdr+Z2eorC1QjbxwW+LbiIChiu/wSVRY6/9p41QbXEsBoIEESPD8JIfxGM/HFRhR miVN2LGiUX8+501piXq4JRrBz+VrQ1HXpDKQj4+MAipVPqL4jsQhbbOQ/OvZa3uA18oA9FXpsC4 lsiVApduLOlK8OxJPvygtB1gMlSisOHLWiZGWjY7uEu9CvW9LPlcz4MzHhIg6/Vemj5KyT/Mcn/ 874K8LgZG8eTnzBac3nvrs12k5BnMUa/4JoBUOjqrJUe4cckYPIjw2sw9rUFqfOqh9LiG76dCFv gBOKqZU3ZrEGT7/yE+VlMkHV+rON26+XWIet5nGZobPXZFm9gAgIkAav3b//X4= Origin: 127 .0.0.1

2.1.2 Make REST API Call

2.1.2.1 API Endpoint

Use the following URL as the API service endpoint for the chosen environment:

Site	SSL	HTTP Method	Host	Port	Context
Sandbox	True	GET	openapi.m-pesa.com	443	/sandbox/ipg/v2/[market]/getSession/
OpenAPI	True	GET	openapi.m-pesa.com	443	/openapi/ipg/v2/[market]/getSession/

Replace the [market] and [currency] parameters with the appropriate values from the below table:

2.1.2.2 Markets

Description	URL Context Value	input_Country Value	input_Currency Value
Vodafone Ghana	vodafoneGHA	GHA	GHS
Vodacom Tanzania	vodacomTZN	TZN	TZS
Vodacom Lesotho	vodacomLES	LES	LSL
Vodacom DR Congo	vodacomDRC	DRC	USD
Vodacom Mozambique	vodacomMOZ	MOZ	MZN

2.1.2.3 API Setup Example

GET <https://openapi.m-pesa.com/sandbox/ipg/v2/vodafoneGHA/getSession/>

2.1.3 API Request

No request parameters are needed for the body of the Generate SessionKey call.

2.1.4 API Synchronous Response

After initiating the API call, the system will respond with the following items:

Parameter Name	Parameter Description	Parameter Possible Values
output_ResponseCode	The result code for the transaction.	INS-0
output_ResponseDesc	The result description for the transaction.	Request processed successfully
output_SessionID	The SessionKey that can be used to call other APIs.	AAA6d1f93910052de0534912jbsj1j2kk

2.1.5 Response Codes

HTTP StatusCode	Response Code	Response Description
200	INS-0	Request processed successfully
400	INS-989	Session Creation Failed

2.1.6 Sample Packets

Sample request and response packets. This API is only available in an end-to-end synchronous flow.

No Request Parameters

2.1.7 Sample Code Snippets

2.1.7.1 Libraries Download

Libraries to assist with integrations.

2.1.7.2 APIKey Encryption

This code snippet can be used when you need to only perform APIKey or SessionKey encryption before attempting any API calls and you would like to use your own HTTP and API handlers to perform the API calls. **Please ensure that you have downloaded the latest library for your coding language above before attempting to use the below code section**

```
using System; using System.Collections.Generic; using System.Threading; using PortalSDK; namespace dotnet { class Program { static void Main(string[] args) { // Public key on the API listener used to encrypt keys String publicKeyKey = "MIICIJANBgqhkiG9w0BAQEFAAOCAg8AMIICGKCAgEArv9yxA69XQKBo24BaF/D+fv1qmGdYjqLQ5WtNBb5tquqGvAvG3WMFETVUSow/LizQalxj2ElMVRUmzu5mGGkxK08bWEXF7a1DE
```



```

vtVJs6nppIlFJc2SnrU14AOrIrB28ogm58JjAl5BOQawOXD5dfSk7MaAA82pVHoIqEu0FxABOK
U+RGtIhRU+ptwlj4bsAJYiPbSX6i7lgefPvwHPYamM0bfI4CmlsUUR3KvCG24rB6FNPcRBhM3jDu
v8ae2kC33w9hEq8qNB55uw51vK7hyXoAa+U7IqPly6nBdlN25gkxEA8yrs1l678cspeXr+3ciRy
qoRgj9RD/ONbJhhxFvt1cLBh+qwK2eqISfBb06eRnNeC7loBokDm3zyCnkOtMDG17IvnMfZfEPF
Cfg5QgJVklmsPpRvQxmEsR9MQRyFVZgy2CWNib7c+jPapYrNwoUbANlN8adU1m6yOuoX7F49x+
OjiG2se0EJ6nafeKUXw/+hiJZvELUYgzKUTMAZVTNZfT8jjb58j8GVtuS+6TM2AutbejaCV84ZK
58E2CRJqhmjQibEU06KpD7oTlEkFy52Y1u0OBXgYpqMzufNPmfdqqqSM4dU70PO8ogyKGiLAIx
CetMj6FCMEA3Kc8K0Ig7/XtFm9By6VxTJK1Mg36TlHaZKP6VzVLXmtesJECaWEEAQ==" ; //
Create Context with API to request a Session ID APIContext context = new AP
IContext(); // Public key context.setPublicKey(publicKey); // Api key conte
xt.setApiKey("6bc4157dbowkdd409118e0978dc6991a"); // Create a request objec
t APIRequest request = new APIRequest(context); // Generate BearerToken Str
ing token = request.createBearerToken(); Console.WriteLine(token); } } }

```

2.1.7.3 Generate SessionKey

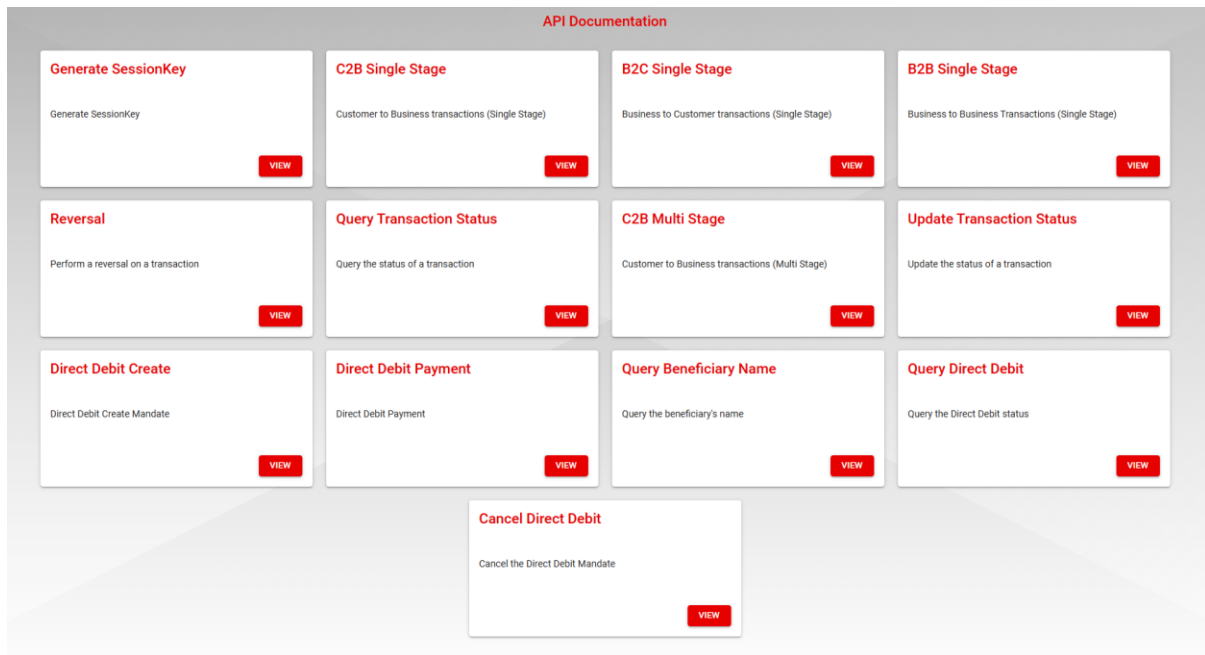
This API should always be the first call to be used as it will return the SessionKey that needs to be used in conjunction with all of the other API calls from the API documentation. **Please ensure that you have downloaded the latest library for your coding language above before attempting to use the below API section**

```

using System; using System.Collections.Generic; using System.Threading; usi
ng PortalSDK; namespace dotnet { class Program { static void Main(string[]
args) { // Public key on the API listener used to encrypt keys String publi
cKey = "MIICIjANBgkqhkiG9w0BAQEFAAOCAg8AMIICCgKCAgEArv9yxA69XQKBo24BaF/D+fv
lqmGdYjqLQ5WtNBb5tquqGvAvG3WMFETVUSow/LizQalxj2ElMvUmzu5mGGkxK08bWEXF7a1DE
vtVJs6nppIlFJc2SnrU14AOrIrB28ogm58JjAl5BOQawOXD5dfSk7MaAA82pVHoIqEu0FxABOK
U+RGtIhRU+ptwlj4bsAJYiPbSX6i7lgefPvwHPYamM0bfI4CmlsUUR3KvCG24rB6FNPcRBhM3jDu
v8ae2kC33w9hEq8qNB55uw51vK7hyXoAa+U7IqPly6nBdlN25gkxEA8yrs1l678cspeXr+3ciRy
qoRgj9RD/ONbJhhxFvt1cLBh+qwK2eqISfBb06eRnNeC7loBokDm3zyCnkOtMDG17IvnMfZfEPF
Cfg5QgJVklmsPpRvQxmEsR9MQRyFVZgy2CWNib7c+jPapYrNwoUbANlN8adU1m6yOuoX7F49x+
OjiG2se0EJ6nafeKUXw/+hiJZvELUYgzKUTMAZVTNZfT8jjb58j8GVtuS+6TM2AutbejaCV84ZK
58E2CRJqhmjQibEU06KpD7oTlEkFy52Y1u0OBXgYpqMzufNPmfdqqqSM4dU70PO8ogyKGiLAIx
CetMj6FCMEA3Kc8K0Ig7/XtFm9By6VxTJK1Mg36TlHaZKP6VzVLXmtesJECaWEEAQ==" ; //
Create Context with API to request a Session ID APIContext context = new AP
IContext(); // Public key context.setPublicKey(publicKey); // Api key conte
xt.setApiKey("6bc4157dbowkdd409118e0978dc6991a"); // Use ssl/https context.
setSsl(true); // Method type (can be GET/POST/PUT) context.setMethodType(AP
IMethodTypes.GET); // API address context.setAddress("openapi.m-pesa.com");
// API Port context.setPort(443); // API Path context.setPath("/sandbox/ipg
/v2/vodafoneGHA/getSession/"); // Parameters can be added to the call as we
ll that on POST will be in JSON format and on GET will be URL // parameters
// context.AddParameter("key", "value"); // Add/update headers context.addH
eader("Origin", "*"); // Create a request object APIRequest request = new A
PIRequest(context); // Do the API call and put result in a response packet
APIResponse response = null; try { response = request.execute(); } catch (
Exception e) { Console.WriteLine("Call failed: {0}", e.Message); } if (resp
onse == null) { throw new Exception("SessionKey call failed to get result.
Please check."); } // Display results Console.WriteLine(response.getStatusC
ode()); foreach(KeyValuePair<String, String> header in response.getHeaders(
)) { Console.WriteLine(header.Key + ": " + header.Value); } foreach(KeyValu
ePair<String, String> item in response.getBody()) { Console.WriteLine(item.
Key + ": " + item.Value); } } } }

```

The image below is full set of MPESA API products



2.2 Customer to Business APIs

2.2.1 API Description

The C2B API call is used as a standard customer-to-business transaction. Funds from the customer's mobile money wallet will be deducted and be transferred to the mobile money wallet of the business. To authenticate and authorize this transaction, M-Pesa Payments Gateway will initiate a USSD Push message to the customer to gather and verify the mobile money PIN number. This number is not stored and is used only to authorize the transaction.

2.2.2 Make REST API Call

2.2.2.1 API Endpoint

Use the following URL as the API service endpoint for the chosen environment:

Site	SSL	HTTP Method	Host	Port	Context
Sandbox	True	POST	openapi.m-pesa.com	443	/sandbox/ipg/v2/[market]/c2bPayment/singleStage/

OpenAPI	True	POST	openapi.m-pesa.com	443	/openapi/ipg/v2/[market]/c2bPayment/singleStage/
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Replace the [market] and [currency] parameters with the appropriate values from the below table:

2.2.2.2 API Markets

Description	URL Context Value	input_Country Value	input_Currency Value
Vodafone Ghana	vodafoneGHA	GHA	GHS
Vodacom Tanzania	vodacomTZN	TZN	TZS
Vodacom Lesotho	vodacomLES	LES	LSL
Vodacom DR Congo	vodacomDRC	DRC	USD
Vodacom Mozambique	vodacomMOZ	MOZ	MZN

2.2.2.3 API Setup Example

POST <https://openapi.m-pesa.com/sandbox/ipg/v2/vodafoneGHA/c2bPayment/singleStage/>

2.2.3 API Headers

The header of the API call makes use of the Session Key received from the Generate SessionKey API call when authorising and authenticating the calling party.

Header Name	Header Value	
Content-Type	application/json	The content-type of the API
Authorization	Bearer [encrypted-SessionKey]	Bearer token authorization is used. The encrypted SessionKey need
Origin	*	Used to limit the origin of the API caller to specific systems. Use the

Content-Type: application/json Authorization: Bearer K2sf0ne61AocJqWx03kTHM BmLl0fQyBQWV7GfH3z1kMxV qMCB7sMXn+CKigIPiawHoNr2IrGdjBdFugyHNxRnGkrfSDkC9PF BhR/D9ePdw15jgggnG8FItGpw4TEwECgzk0sEmYmKaB9K6lgvs4rs7TMPpTom7uX72+tZ+qMqRV t7LrGsngBrkXjbOxc19fkImVHjehzjVCOSOsROsx4nRlM9wtl4KSLgKCBzmLePRmwu/IOCF3NrI QLVYYqUDDSGYbHURk1L6d4kxJffFM/wa2BPoeuG59gcglDyEaQ5Sv5VnaGKDh1XTBrvKS+J9Efw g7s1CWLQxhtYdr+Z2eorC1QjbxwW+LbiIChiu/wSVRY6/9p41QbXESBoIEESPD8JIfxGM/HFRhR miVN2LGiuX8+501piXq4JRrBz+VrQ1HXpDKQj4+MAipVPqL4jsQhbbOQ/OvZa3uA18oA9FXpsC4 1siVApduL0lK8OxJPvygtB1gMlSisOHLWiZGWjY7uEu9CvW9LP1cz4MzHhIg6/Vemj5KyT/Mcn/ 874K8LgZG8eTnzBac3nvrs12k5BnMUa/4JoBU0jqRJuE4cckYPIjw2sw9rUFqfOqh9LiG76dCFv gBOKqZU3ZrEGT7/yE+VlMkHV+rON26+XWIet5nGZobPXZFm9gAgIkAav3b//X4= Origin: 127 .0.0.1

2.3 API Requests

The body of the API call is made up of the following parameters and values:

Parameter Name	Parameter Description	Parameter Mandatory	Parameter Regex Validation	Parameter Possible Values
input_Amount	The transaction amount. This amount will be moved from the organization's account to the customer's account.	True	^\d*\.\?\d+\$	10.00
input_CustomerMSISDN	The MSISDN of the customer where funds will be debitted from.	True	^[0-9]{12,14}\$	254707161122
input_Country	The country of the mobile money platform where the transaction needs happen on.	True	N/A	GHA
input_Currency	The currency in which the transaction should take place.	True	^[a-zA-Z]{1,3}\$	GHS
input_ServiceProviderCode	The shortcode of the organization where funds will be credited to.	True	^([0-9A-Za-z]{4,12})\$	ORG001
input_TransactionReference	The customer's transaction reference	True	^[0-9a-zA-Z\w]{1,20}\$	T12344C
input_ThirdPartyConversationID	The third party's transaction reference on their system.	True	^[0-9a-zA-Z\w]{1,40}\$	1e9b774d1da34af78412a498cbc28f5e
input_PurchasedItemsDesc	Description of purchased items	True	^[0-9a-zA-Z\w]{1,256}\$	Handbag, Black shoes

2.4 API Responses

The response back to the application can either be synchronous or asynchronous, depending the business use case. Synchronous responses will make use of the same session to return the results. For asynchronous responses, the user will need to build a listener to receive the asynchronous response once the transaction has finalised on the payment system.

The response back to the application can either be synchronous or asynchronous, depending the business use case. Synchronous responses will make use of the same session to return the results. For asynchronous

responses, the user will need to build a listener to receive responses once the transaction has finalised on the walletting system.

2.4.1 Synchronous Responses

Parameter Name	Parameter Description	Parameter Possible Values
output_ResponseCode	The result code for the transaction.	INS-0
output_ResponseDesc	The result description for the transaction.	Request processed successfully
output_TransactionID	The transaction identifier that gets generated on the Mobile Money platform. This is used to query transactions on the Mobile Money Platform.	hv9ahxcg4ccv
output_ConversationID	The OpenAPI platform generates this as a reference to the transaction.	fd1e9143d22544459f7c66e1860ef276
output_ThirdPartyConversationID	The incoming reference from the third party system. When there are queries about transactions, this will usually be used to track a transaction.	1e9b774d1da34af78412a498cbc28f5e

2.4.2 Asynchronous Responses

Once the transaction is initiated, the OpenAPI will respond synchronously with the following information before the session is closed.

Parameter Name	Parameter Description	Parameter Possible Values
output_ResponseCode	The result code for the transaction.	INS-0
output_ResponseDesc	The result description for the transaction.	Request processed successfully
output_ConversationID	The OpenAPI platform generates this as a reference to the transaction.	fd1e9143d22544459f7c66e1860ef276

output_ThirdPartyConversationID	The incoming reference from the third party system. When there are queries about transactions, this will usually be used to track a transaction.	1e9b774d1da34af78412a498cbc28f5e
---------------------------------	--	----------------------------------

The Conversation ID can be used in a Query Transaction Status call to determine the status of the process flow. Once the transaction has been completed, the OpenAPI will provide the result of the transaction in an asynchronous message to the value listed in the Response URL of the request parameter in the body. Please note that the developed application will need to have a separate listener to receive the result.

Parameter Name	Parameter Description	Parameter Possible Values
input_OriginalConversationID	The OpenAPI platform generates this as a reference to the transaction.	fd1e9143d22544459f7c66e1860ef276
input_TransactionID	The transaction identifier that gets generated on the Mobile Money platform. This is used to query transactions on the Mobile Money Platform.	hv9ahxcg4ccv
input_ResultCode	The result code for the transaction.	INS-0
input_ResultDesc	The result description for the transaction.	Request processed successfully
input_ThirdPartyConversationID	The incoming reference from the third party system. When there are queries about transactions, this will usually be used to track a transaction.	1e9b774d1da34af78412a498cbc28f5e

It is expected that the listener confirms receipt of the result by responding the below content which closes out the session:

Parameter Name	Parameter Description	Parameter Possible Values
output_OriginalConversationID	The OpenAPI platform generates this as a	fd1e9143d22544459f7c66e1860ef276

	reference to the transaction.	
output_ResponseCode	The result code for the transaction.	0
output_ResponseDesc	The result description for the transaction.	Successfully Accepted Result
output_ThirdPartyConversationID	The incoming reference from the third party system. When there are queries about transactions, this will usually be used to track a transaction.	1e9b774d1da34af78412a498cbc28f5e

2.4.3 Response Codes

HTTP StatusCode	Response Code	Response Description
201	INS-0	Request processed successfully
400	INS-1	Internal Error
401	INS-6	Transaction Failed
408	INS-9	Request timeout
409	INS-10	Duplicate Transaction
400	INS-13	Invalid Shortcode Used
400	INS-15	Invalid Amount Used
400	INS-17	Invalid Transaction Reference. Length Should Be Between 1 and 20.
400	INS-20	Not All Parameters Provided. Please try again.
400	INS-21	Parameter validations failed. Please try again.
400	INS-26	Invalid Currency Used
400	INS-28	Invalid ThirdPartyConversationID Used
400	INS-30	Invalid Purchased Items Description Used
400	INS-990	Customer Transaction Value Limit Breached
400	INS-991	Customer Transaction Count Limit Breached
400	INS-992	Multiple Limits Breached

400	INS-993	Organization Transaction Count Limit Breached
400	INS-994	Organization Transaction Value Limit Breached
400	INS-995	API Single Transaction Limit Breached
400	INS-996	API Being Used Outside Of Usage Time
400	INS-997	API Not Enabled
400	INS-998	Invalid Market
422	INS-2006	Insufficient balance
400	INS-2051	MSISDN invalid.

2.5 Sample Packets

Sample request and response packets. If you want to make use of a end to end synchronous flow, use the following tabs: Request (Thirdparty -> OpenAPI) Sync Flow Response (OpenAPI -> Thirdparty) If you want to make use of a end to end asynchronous flow, use the following tabs: Request (Thirdparty -> OpenAPI) Async Flow Response (OpenAPI -> Thirdparty) OpenAPI Async Flow Request (OpenAPI -> Thirdparty) ThirdParty Async Flow Response (Thirdparty -> OpenAPI)

```
{ "input_Amount": "10", "input_Country": "GHA", "input_Currency": "GHS", "input_CustomerMSISDN": "000000000001", "input_ServiceProviderCode": "000000", "input_ThirdPartyConversationID": "asv02e5958774f7ba228d83d0d689761", "input_TransactionReference": "T1234C", "input_PurchasedItemsDesc": "Shoes" }
```

2.6 Sample Code Snippets

2.6.1 Libraries Download

Libraries to assist with integrations.

2.6.2 C2B Single Stage

This API should always be used in conjunction with a valid SessionKey. Please refer to the Generate SessionKey API documentation page to see how to generate a SessionKey or see the below combined API code snippet where the Generate SessionKey API is combined with the C2B Single Stage API. **Please ensure that you have downloaded the latest library for your coding language above before attempting to use the below API section**

```
using System; using System.Collections.Generic; using System.Threading; using PortalSDK; namespace dotnet { class Program { static void Main(string[] args) { // Public key on the API listener used to encrypt keys String publi
```



```
cKey = "MIICIjANBgkqhkiG9w0BAQEFAAOCAg8AMIICCgKCAGEarv9yxA69XQKBo24BaF/D+fv
lqmGdYjqLQ5WtNBb5tquqGvAvG3WMFETVUSow/LizQalxj2ElMvUmzu5mGGkxK08bWEXF7a1DE
vtVJs6nppIlfJc2SnrU14AOIrB28ogm58JjAl5BOQawOXD5dfSk7MaAA82pVHoIqEu0FxA8BOK
U+RGTiHRU+ptwlj4bsAJYiPbSX6i7lGfPvwHPYamM0bfI4CmlsUUR3KvCG24rB6FNPCRBhM3jDu
v8ae2kC33w9hEq8qNB55uw51vK7hyXoAa+U7IqPly6nBdlN25gkxEA8yrs11678cspeXr+3ciRy
qoRgj9RD/ONbJhhxFvt1cLBh+qwK2eqISfBb06eRnNeC71oBokDm3zyCnkOtMDG17IvnMfZfEPF
Cfg5QgJVklmsPpRvQxmEsR9MQRyFVzgy2CWNib7c+jPapyrNwoUbANlN8adUlm6yOuoX7F49x+
OjiG2se0EJ6nafeKUXw/+hiJZvELUYgzKUTMAZVTNZft8jjb58j8GVtuS+6TM2AutbejaCV84ZK
58E2CRJqhmjQibEU06KPdD7oTlEkFy52Y1uOOBXgYpqMzufNPmfdqqqSM4dU70PO8ogyKGiLAIx
CetMjjm6FCMEA3Kc8K0Ig7/XtFm9By6VxTJK1Mg36TlHaZKP6VzVLXMtesJECawEAAQ=="; //
Create Context with API to request a Session ID APIContext context = new AP
IContext(); // Public key context.setPublicKey(publicKey); // Session key c
ontext.setApiKey("6bc4157dbowkdd409118e0978dc6991a"); // Use ssl/https cont
ext.setSsl(true); // Method type (can be GET/POST/PUT) context.setMethodTyp
e(APIMethodTypes.POST); // API address context.setAddress("openapi.m-pesa.c
om"); // API Port context.setPort(443); // API Path context.setPath("/sandb
ox/ipg/v2/vodafoneGHA/c2bPayment/singleStage/"); // Parameters can be added
to the call as well that on POST will be in JSON format and on GET will be
URL // parameters // context.addParameter("key", "value"); context.addParam
eter("input_Amount", "10"); context.addParameter("input_Country", "GHA"); c
ontext.addParameter("input_Currency", "GHS"); context.addParameter("input_C
ustomerMSISDN", "000000000001"); context.addParameter("input_ServiceProvide
rCode", "000000"); context.addParameter("input_ThirdPartyConversationID", "
asv02e5958774f7ba228d83d0d689761"); context.addParameter("input_Transaction
Reference", "T1234C"); context.addParameter("input_PurchasedItemsDesc", "Sh
oes"); // Add/update headers context.addHeader("Origin", "*"); // Create a
request object APIRequest request = new APIRequest(context); // Do the API
call and put result in a response packet APIResponse response = null; try {
response = request.execute(); } catch (Exception e) { Console.WriteLine("C
all failed: {0}", e.Message); } if (response == null) { throw new Exception
("API call failed to get result. Please check."); } // Display results Cons
ole.WriteLine(response.getStatusCode()); foreach(KeyValuePair<String, Strin
g> header in response.getHeaders()) { Console.WriteLine(header.Key + ": " +
header.Value); } foreach(KeyValuePair<String, String> item in response.getB
ody()) { Console.WriteLine(item.Key + ": " + item.Value); } } }
```

2.6.3 C2B Single Stage Combined

Please ensure that you have downloaded the latest library for your coding language above before attempting to use the below API section

```
using System; using System.Collections.Generic; using System.Threading; usi
ng PortalSDK; namespace dotnet { class Program { static void Main(string[]
args) { // Public key on the API listener used to encrypt keys String publi
cKey = "MIICIjANBgkqhkiG9w0BAQEFAAOCAg8AMIICCgKCAGEarv9yxA69XQKBo24BaF/D+fv
lqmGdYjqLQ5WtNBb5tquqGvAvG3WMFETVUSow/LizQalxj2ElMvUmzu5mGGkxK08bWEXF7a1DE
vtVJs6nppIlfJc2SnrU14AOIrB28ogm58JjAl5BOQawOXD5dfSk7MaAA82pVHoIqEu0FxA8BOK
U+RGTiHRU+ptwlj4bsAJYiPbSX6i7lGfPvwHPYamM0bfI4CmlsUUR3KvCG24rB6FNPCRBhM3jDu
v8ae2kC33w9hEq8qNB55uw51vK7hyXoAa+U7IqPly6nBdlN25gkxEA8yrs11678cspeXr+3ciRy
qoRgj9RD/ONbJhhxFvt1cLBh+qwK2eqISfBb06eRnNeC71oBokDm3zyCnkOtMDG17IvnMfZfEPF
Cfg5QgJVklmsPpRvQxmEsR9MQRyFVzgy2CWNib7c+jPapyrNwoUbANlN8adUlm6yOuoX7F49x+
OjiG2se0EJ6nafeKUXw/+hiJZvELUYgzKUTMAZVTNZft8jjb58j8GVtuS+6TM2AutbejaCV84ZK
58E2CRJqhmjQibEU06KPdD7oTlEkFy52Y1uOOBXgYpqMzufNPmfdqqqSM4dU70PO8ogyKGiLAIx
CetMjjm6FCMEA3Kc8K0Ig7/XtFm9By6VxTJK1Mg36TlHaZKP6VzVLXMtesJECawEAAQ=="; //
Create Context with API to request a Session ID APIContext context = new AP
IContext(); // Public key context.setPublicKey(publicKey); // Api key conte
xt.setApiKey("6bc4157dbec34d409118e0978dc6dd17"); // Use ssl/https context.
setSsl(true); // Method type (can be GET/POST/PUT) context.setMethodType(AP
IMethodTypes.GET); // API address context.setAddress("openapi.m-pesa.com");
// API Port context.setPort(443); // API Path context.setPath("/sandbox/ipg
/v2/vodafoneGHA/getSession/"); // Parameters can be added to the call as we
```

```

11 that on POST will be in JSON format and on GET will be URL // parameters
// context.AddParameter("key", "value"); // Add/update headers context.addHeader("Origin", ""); // Create a request object APIRequest request = new APIRequest(context); // Do the API call and put result in a response packet
APIResponse response = null; try { response = request.execute(); } catch (Exception e) { Console.WriteLine("Call failed: {0}", e.Message); } if (response == null) { throw new Exception("SessionKey call failed to get result. Please check."); } // Display results Console.WriteLine(response.StatusCode()); foreach(KeyValuePair<String, String> header in response.getHeaders()) { Console.WriteLine(header.Key + ": " + header.Value); } foreach(KeyValuePair<String, String> item in response.getBody()) { Console.WriteLine(item.Key + ": " + item.Value); } String sessionId = response.getBody()["output_SessionID"]; // The above call issued a sessionID which will be used as the API key in calls that needs the sessionID context = new APIContext(); context.setPublicKey(publicKey); context.setApiKey(sessionId); context.setSsl(true); context.setMethodType(APIMethodTypes.POST); context.setAddress("openapi.m-pesa.com"); context.setPort(443); context.setPath("/sandbox/ipg/v2/vodafoneGHA/c2bPayment/singleStage/"); context.AddParameter("input_Amount", "10"); context.AddParameter("input_Country", "GHA"); context.AddParameter("input_Currency", "GHS"); context.AddParameter("input_CustomerMSISDN", "0000000000001"); context.AddParameter("input_ServiceProviderCode", "000000"); context.AddParameter("input_ThirdPartyConversationID", "asv02e5958774f7ba228d83d0d689761"); context.AddParameter("input_TransactionReference", "T1234C"); context.AddParameter("input_PurchasedItemsDesc", "Shoes"); context.addHeader("Origin", ""); request = new APIRequest(context); // SessionID can take up to 30 seconds to become 'live' in the system and will be invalid until it is Thread.Sleep(30 * 1000); response = null; try { response = request.execute(); } catch (Exception e) { Console.WriteLine("Call failed: {0}", e.Message); } Console.WriteLine(response.StatusCode()); foreach(KeyValuePair<String, String> header in response.getHeaders()) { Console.WriteLine(header.Key + ": " + header.Value); } foreach(KeyValuePair<String, String> item in response.getBody()) { Console.WriteLine(item.Key + ": " + item.Value); } } }

```